
RESEARCH PAPER

Society Under Monetary Stress VIII: The Confidence Ledger

Inflation Volatility, Expectations, and Consumer Sentiment

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Abstract

This paper revises the confidence-ledger analysis as a descriptive consumer-sentiment paper rather than an institutional-trust paper. Using public FRED data, it combines monthly University of Michigan consumer sentiment and expected inflation with realized inflation, rolling inflation volatility, unemployment, real disposable-income growth, gasoline inflation, policy rates, debt service, and revolving credit. The evidence is not causal. It shows that sentiment is lower when expected inflation and inflation volatility are elevated, and the relationship remains visible outside the 2020-2022 shock window. Monthly correlations and standardized conditional associations strengthen the descriptive foundation relative to annual milestones, while preserving the core caveat: Michigan expected inflation and Michigan sentiment share a survey environment and may reflect common pessimism.

JEL codes: D12, D14, D84, E31 **Keywords:** consumer sentiment, inflation expectations, inflation volatility, household expectations, planning reliability

1. Introduction

Consumer confidence is not reducible to the unemployment rate. Households can be employed and still feel that the future is difficult to plan. Prices may change too quickly. Expected inflation may rise. Real income may weaken. Saving may feel thin. Credit and interest-rate conditions may make large commitments less comfortable. This paper calls the household-facing record of those signals the **confidence ledger**.

The phrase is a measurement frame, not a formal theory. A stronger microdata paper would model how households convert inflation experiences, balance-sheet exposure, labor risk, and news into beliefs and behavior. This paper is more modest. It asks whether a clean public-data ledger of sentiment, inflation expectations, inflation volatility, labor conditions, income growth, rates, and credit context describes the main confidence episodes better than unemployment alone.

An external-style critique of the earlier version raised several valid issues. Annual correlations left too little information unused. The figures overlaid variables with incompatible scales. The paper leaned on selected milestone years. It needed statistical uncertainty, robustness outside the pandemic/inflation-shock period, and a clearer warning that Michigan sentiment and Michigan expected inflation come from the same survey environment. This revision addresses those issues while keeping the public-data scope.

The paper is now framed around **consumer sentiment**, not institutional trust. Direct institutional trust would require General Social Survey, Gallup, ANES, or similar repeated cross-sectional microdata. That remains a future extension. The current paper contributes a transparent descriptive foundation: monthly co-movement, uncertainty intervals, robustness outside 2020-2022, standardized conditional associations, and clearer charts.

2. Data and measurement

The analysis uses public FRED CSV downloads. The main monthly panel runs from January 1978 through December 2025 because University of Michigan expected inflation (MICH) begins in 1978 and consumer sentiment (UMCSENT) is available over the full period. The annual milestone table begins in 1994 only because U-6 underemployment is first available then; the monthly statistical tables use the longer available sample where possible.

The main outcome is University of Michigan consumer sentiment (UMCSENT). The central expectations variable is University of Michigan expected inflation over the next year (MICH). The fact that both series come from the same survey ecosystem is a limitation. Their relationship may reflect inflation expectations specifically, but it may also reflect common respondent mood, survey framing, political salience, media attention, or generalized pessimism. The paper therefore treats MICH as part of the confidence ledger, not as an independent causal shock.

Macroeconomic context includes CPI (CPIAUCSL), unemployment (UNRATE), U-6 underemployment (U6RATE), effective federal funds rate (FEDFUNDS), 10-year Treasury yield (DGS10), real disposable personal income (DSPIC96), saving rate (PSAVERT), household debt service (TDSP), revolving consumer credit (REVOLSL), disposable personal income (DSPI), regular gasoline prices (GASREGW), and the NBER recession indicator (USREC).

Constructed variables are:

- monthly year-over-year CPI inflation;
- rolling 12-month volatility of monthly CPI inflation;
- realized inflation less prior-year Michigan expected inflation;
- year-over-year gasoline inflation;
- year-over-year real disposable-income growth;
- revolving credit divided by disposable personal income, after converting REVOLSL from millions to billions.

The paper reports correlations with Fisher-z confidence intervals and standardized OLS associations. These are not identification strategies. They are included to keep the descriptive claims from leaning too heavily on selected years.

3. Milestones: the confidence ledger in narrative form

Table 1 provides narrative context. The statistical evidence follows in monthly tables.

Table 1. Confidence-ledger milestones

Selected years provide narrative context; statistical tables use the full monthly panel.

Year	Consumer sentiment	Expected inflation	CPI inflation	12m inflation volatility	Gasoline inflation	Unemployment	U-6	Real DPI growth	Saving rate	Fed funds	Debt service/DPI	Revolving credit/DPI
1994	92.3	3.0%	2.6%	0.1%	-2.0%	6.1%	10.8%	2.6%	6.8%	4.2%		6.4%
2000	107.6	3.0%	3.4%	0.1%	27.1%	4.0%	7.0%	4.9%	4.3%	6.2%		8.7%
2007	85.6	3.2%	2.9%	0.2%	4.6%	4.6%	8.3%	2.3%	2.5%	5.0%	15.7%	9.2%
2019	96.0	2.6%	1.8%	0.1%	-3.7%	3.7%	7.1%	3.1%	7.3%	2.2%	11.6%	6.7%
2022	59.0	5.0%	8.0%	0.2%	32.2%	3.6%	6.9%	-5.7%	3.4%	1.7%	10.6%	5.9%
2024	72.5	2.9%	3.0%	0.1%	-0.6%	4.0%	7.5%	2.9%	5.4%	5.1%	11.1%	6.1%
2025	57.6	4.8%	2.7%	0.1%	-2.8%	4.3%	8.0%	1.7%	4.6%	4.2%	11.2%	5.7%

The table shows why unemployment alone is insufficient. In 2022, unemployment averaged 3.6 percent and U-6 averaged 6.9 percent, but sentiment averaged only 59.0. That low-confidence reading coincided with 8.0 percent CPI inflation, 5.0 percent expected inflation, falling real disposable income, high gasoline inflation, and a low saving rate. By 2024, sentiment recovered as inflation and expected inflation fell. In 2025, sentiment weakened again even though realized CPI inflation was much lower than in 2022, because expected inflation rose and real-income growth slowed.

The narrative should not be overstated. These are aggregate co-movements. They do not prove that expected inflation caused sentiment to fall. They do show that the confidence ledger can deteriorate even when labor-market aggregates remain strong.

4. Monthly evidence and uncertainty

Table 2 replaces the earlier version's annual correlation table with monthly correlations, sample sizes, and confidence intervals. It also reports a robustness sample excluding 2020-2022 so that the inflation-shock period does not carry the entire result.

Table 2. Monthly descriptive correlations with consumer sentiment

Pearson correlations using monthly observations. Confidence intervals use Fisher z approximation. These are descriptive co-movements, not causal estimates.

Indicator	Sample	N	r	95% CI
Expected inflation (Michigan)	full	575	-0.50	[-0.56, -0.44]
Expected inflation (Michigan)	ex20202022	539	-0.50	[-0.56, -0.43]
CPI inflation, y/y	full	575	-0.42	[-0.48, -0.35]
CPI inflation, y/y	ex20202022	539	-0.39	[-0.46, -0.32]
12-month inflation volatility	full	570	-0.33	[-0.40, -0.25]
12-month inflation volatility	ex20202022	534	-0.30	[-0.37, -0.22]
Realized inflation less prior-year expected inflation	full	563	-0.20	[-0.28, -0.12]
Realized inflation less prior-year expected inflation	ex20202022	527	-0.12	[-0.21, -0.04]
Gasoline inflation, y/y	full	412	-0.01	[-0.11, 0.09]
Gasoline inflation, y/y	ex20202022	376	0.10	[0.00, 0.20]
Unemployment	full	575	-0.32	[-0.39, -0.25]
Unemployment	ex20202022	539	-0.40	[-0.47, -0.33]
U-6 underemployment	full	383	-0.36	[-0.44, -0.27]
U-6 underemployment	ex20202022	347	-0.46	[-0.54, -0.38]
Real DPI growth, y/y	full	575	0.34	[0.27, 0.41]
Real DPI growth, y/y	ex20202022	539	0.43	[0.35, 0.49]
Personal saving rate	full	575	-0.13	[-0.21, -0.05]
Personal saving rate	ex20202022	539	-0.19	[-0.27, -0.10]
Fed funds rate	full	575	-0.04	[-0.12, 0.04]
Fed funds rate	ex20202022	539	-0.09	[-0.18, -0.01]
Debt service / DPI	full	83	-0.00	[-0.22, 0.21]
Debt service / DPI	ex20202022	71	-0.11	[-0.34, 0.12]
Revolving credit / DPI	full	575	0.26	[0.19, 0.34]
Revolving credit / DPI	ex20202022	539	0.26	[0.18, 0.34]

Expected inflation has the strongest negative bivariate association with sentiment: -0.50 in the full monthly sample and -0.50 excluding 2020-2022. CPI inflation and rolling inflation volatility are also negative. The result therefore does not depend entirely on the 2022 inflation shock, although that episode is an important example.

The same-survey caveat is essential. The Michigan expected-inflation correlation may partly reflect a common sentiment complex rather than a clean causal channel. That is why the table also includes realized inflation, forecast error, unemployment, income growth, and other indicators. The pattern is broad, but not dispositive.

Table 3 reports a compact standardized conditional association. It is not causal, but it helps check whether expected inflation and inflation volatility remain associated with sentiment when labor, income, gasoline inflation, rates, and recession status are included together.

Table 3. Standardized monthly association with consumer sentiment

OLS with standardized variables and intercept. This is not an identification strategy; it is a compact conditional association table.

Variable	Coefficient	SE
Intercept	-0.00	0.03
Expected inflation (Michigan)	-0.67	0.03
12-month inflation volatility	-0.12	0.03
Unemployment	-0.34	0.04
Real DPI growth, y/y	0.12	0.03
Gasoline inflation, y/y	0.22	0.03
Fed funds rate	0.08	0.04
usrec	-0.12	0.03

N = 412 monthly observations.

The coefficient on expected inflation is large and negative in standardized terms. Inflation volatility is also negative. Unemployment is negative, and real disposable-income growth is positive. The gasoline coefficient is positive in this specification, a reminder that conditional correlations can behave unintuitively when variables share cycles, timing, and collinearity. This is exactly why the paper does not claim identification.

5. Figures redesigned around standardized evidence

The earlier version's figures used common axes for variables with incompatible scales. The revision uses standardized time-series charts and a scatterplot.

Figure 1. Sentiment and expected inflation, standardized

Monthly z-scores; expected inflation is inverted visually by its negative co-movement

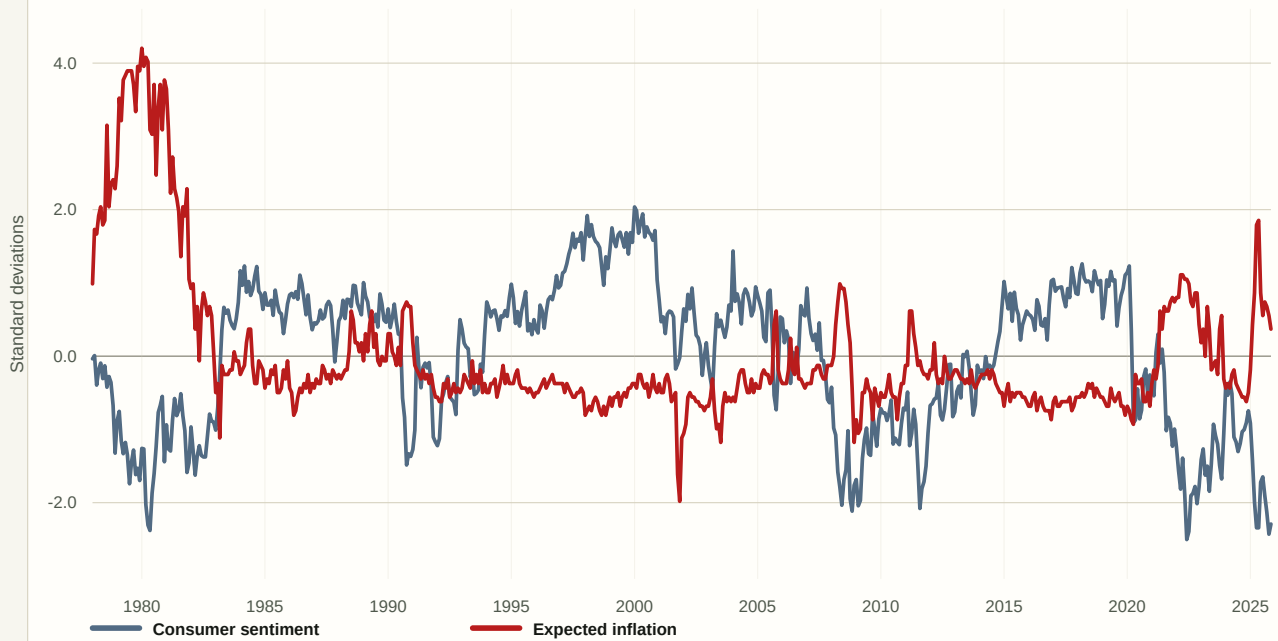


Figure 1. Sentiment and expected inflation, standardized

Figure 1 shows standardized consumer sentiment and expected inflation. The inverse relationship is visible across the full monthly sample, especially during high-inflation-expectation episodes. The figure should still be read with the same-survey caveat.

Figure 2. Sentiment and inflation volatility, standardized

Monthly z-scores; volatility is rolling 12-month monthly CPI inflation volatility

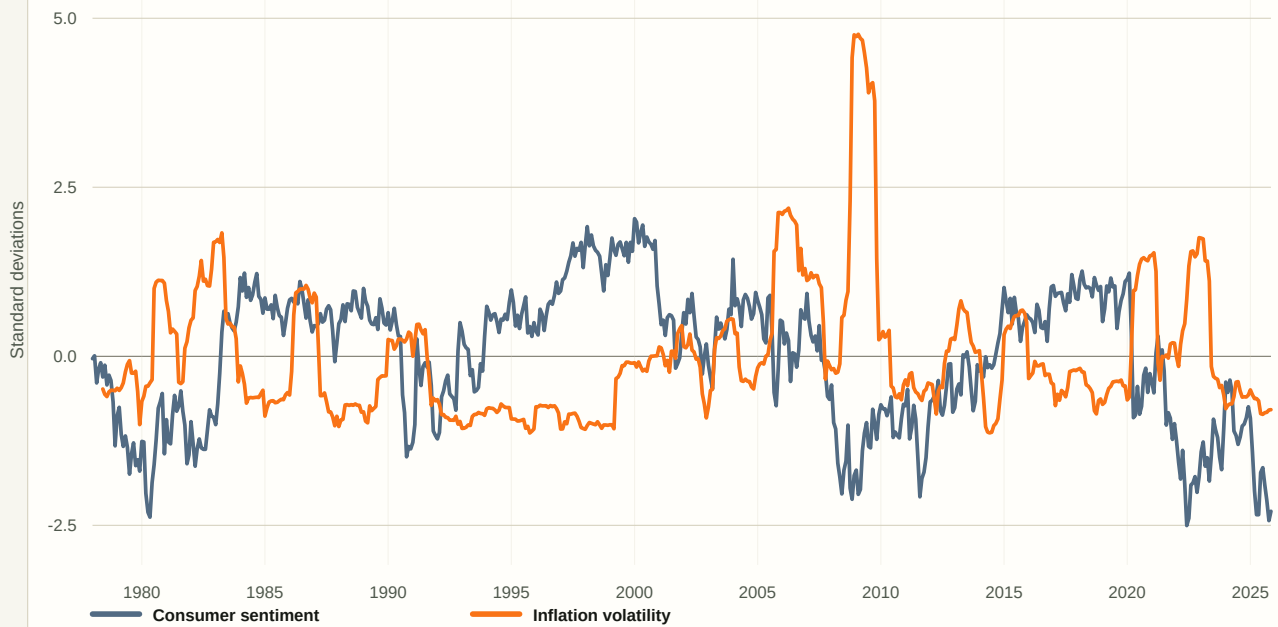


Figure 2. Sentiment and inflation volatility, standardized

Figure 2 shows sentiment and rolling inflation volatility. Volatility is weaker than expected inflation but still moves adversely in low-confidence periods. This supports treating price instability, not only the annual inflation rate, as part of the household planning environment.

Figure 3. Labor and income context, standardized

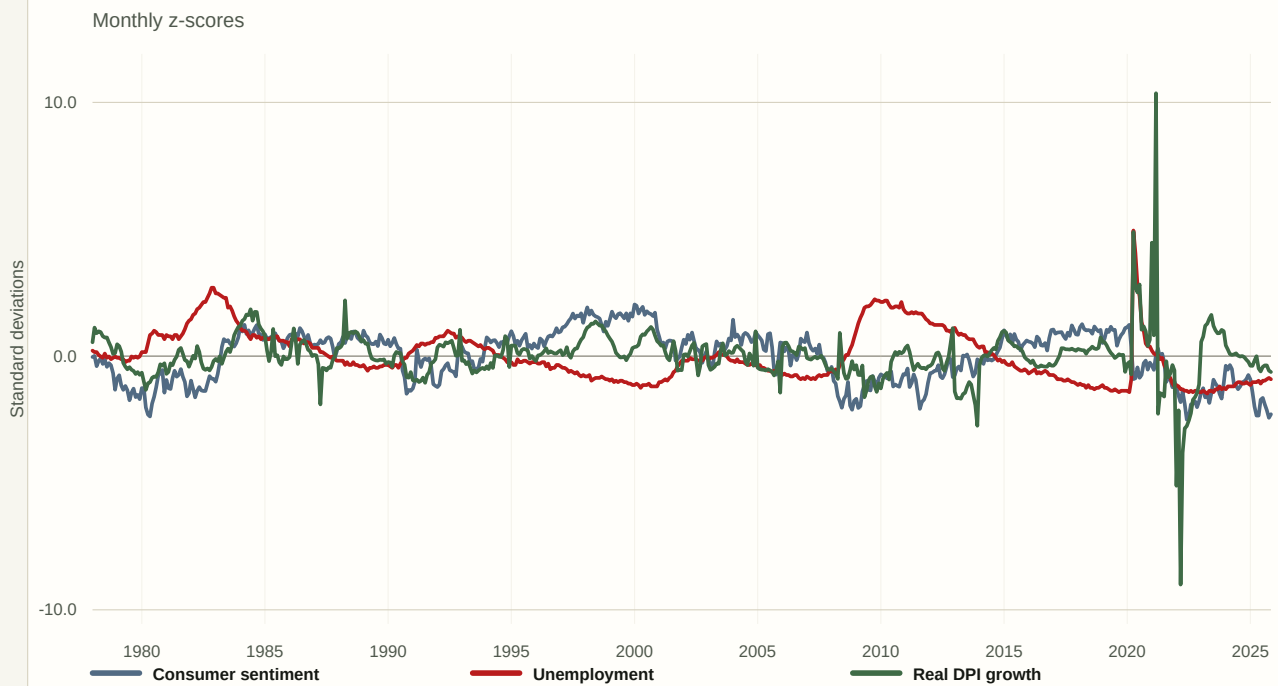


Figure 3. Labor and income context, standardized

Figure 3 shows why labor markets and income growth belong in the ledger. Sentiment falls in labor-market stress periods, but the 2022 episode again stands out because sentiment weakened without a major unemployment spike.

Figure 4. Rate and credit context, standardized

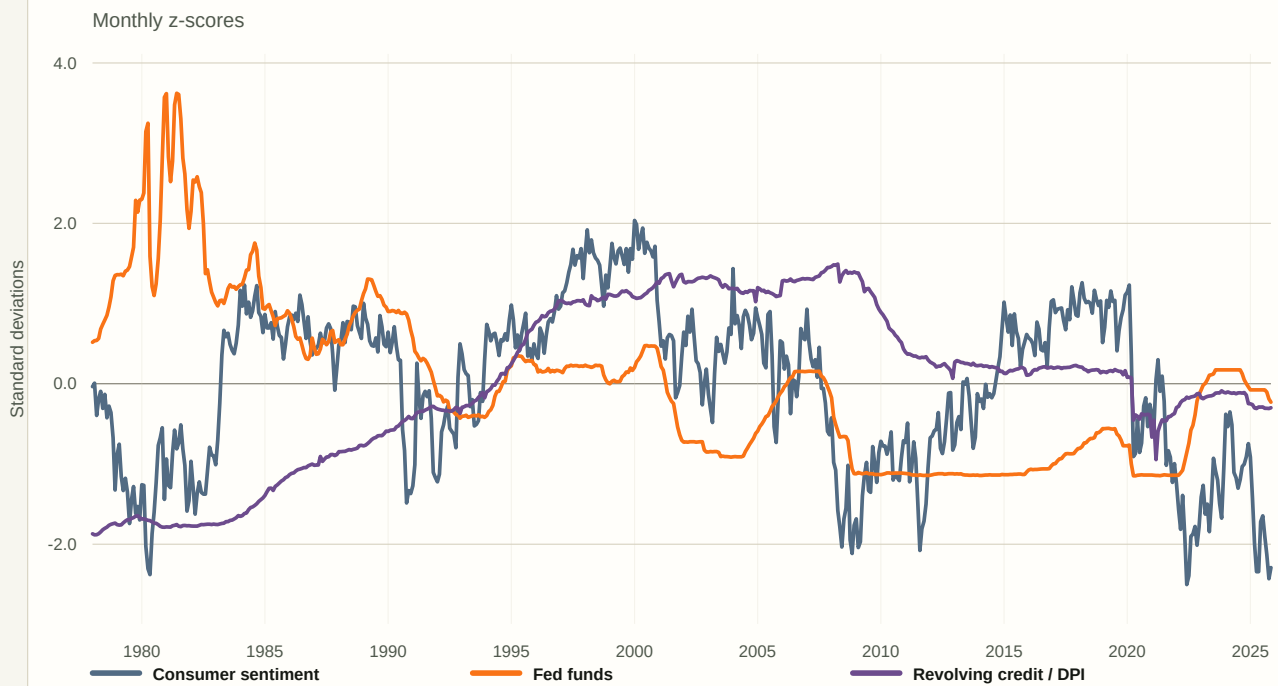


Figure 4. Rate and credit context, standardized

Figure 4 shows rate and revolving-credit context. These variables do not deliver a simple story. In expansions, credit access can rise with sentiment; in stress periods, credit burdens can matter more for constrained households than for the aggregate average. The chart is context, not proof.

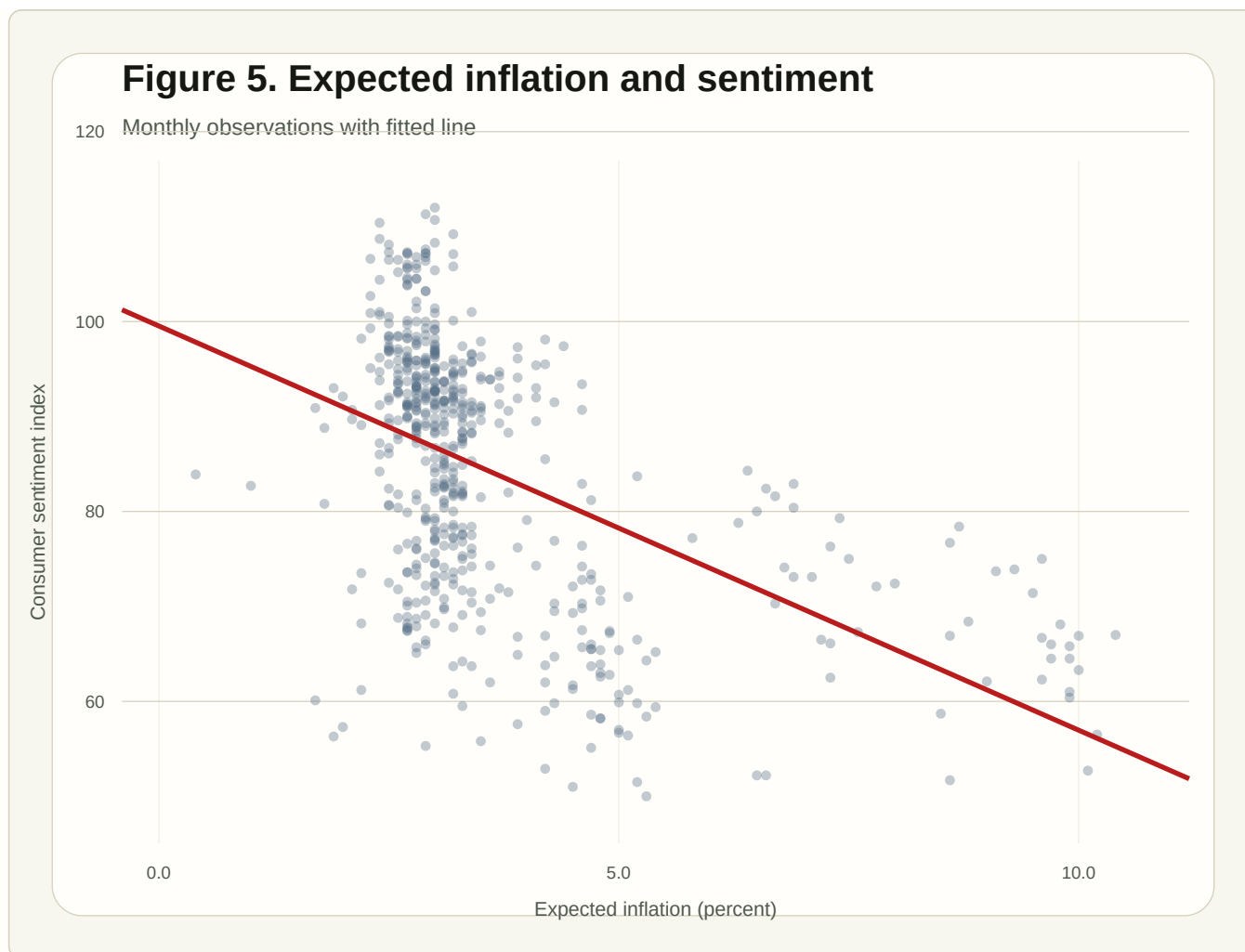


Figure 5. Expected inflation and sentiment

Figure 5 presents the expected-inflation/sentiment relationship as a scatterplot with fitted line. The pattern is negative, but it should not be mistaken for a structural demand curve or policy response function.

6. What this paper adds, and what it does not

The paper's contribution is not that economists should be surprised by the relationship between inflation expectations and sentiment. The contribution is a disciplined public-data framework for separating five ideas that are often blended together: consumer sentiment, inflation expectations, realized inflation, inflation instability, and institutional trust.

That separation matters for public interpretation. A weak confidence reading is not necessarily a labor-market story. It is not necessarily a partisan story. It is not necessarily institutional distrust. It can be a ledger of household-facing planning signals. But this paper cannot determine which signal dominates for which household.

A stronger research version would require microdata. Michigan components or question-level data could separate current personal finances, expected personal finances, expected business conditions, buying conditions, and inflation expectations. Survey of Consumer Expectations data could compare expectation

measures. GSS, Gallup, or ANES data could test institutional confidence directly. Consumption or durable-purchase outcomes could test whether the ledger predicts behavior, not merely sentiment.

7. Conclusion

Consumer confidence is part of monetary stress because it records whether the future feels plannable. The revised evidence is still descriptive, but it is stronger than the first annual version. Monthly public data show that sentiment is lower when expected inflation is higher, realized inflation is higher, inflation is more volatile, unemployment is weaker, and real disposable-income growth is softer. These patterns remain visible outside the 2020-2022 window.

The central caution is equally important. Expected inflation and sentiment share a survey environment. Aggregate public data cannot identify household-level belief formation or institutional trust. This paper should therefore be read as a cleaned descriptive foundation, not a final causal account.

The confidence ledger is useful if it keeps measurement honest. It says: do not infer household confidence from unemployment alone; do not infer institutional trust from consumer sentiment alone; and do not confuse co-movement with mechanism. Under those constraints, the evidence still points to a real public-data pattern: price instability and inflation expectations are central to the way households experience macroeconomic stress.

References

- Carroll, Christopher D., Jeffrey C. Fuhrer, and David W. Wilcox. 1994. "Does Consumer Sentiment Forecast Household Spending? If So, Why?" *American Economic Review* 84 (5): 1397–1408.
- Coibion, Olivier, Yuriy Gorodnichenko, and Michael Weber. 2022. "The Expected, Perceived, and Realized Inflation of U.S. Households before and during the COVID19 Pandemic." *IMF Economic Review* 70: 391–442.
- Curtin, Richard. 2019. *Consumer Expectations: Micro Foundations and Macro Impact*. Cambridge University Press.
- Federal Reserve Bank of St. Louis. FRED Economic Data. Series listed in replication note.
- Kuchler, Theresa, and Basit Zafar. 2019. "Personal Experiences and Expectations about Aggregate Outcomes." *Journal of Finance* 74 (5): 2491–2542.
- Malmendier, Ulrike, and Stefan Nagel. 2016. "Learning from Inflation Experiences." *Quarterly Journal of Economics* 131 (1): 53–87.
- University of Michigan Surveys of Consumers. University of Michigan consumer sentiment and inflation expectations series, distributed through FRED.

Replication note

The analysis uses public FRED CSV downloads pulled in May 2026. Series include UMCSENT, MICH, CPIAUCSL, UNRATE, U6RATE, FEDFUNDS, DGS10, DSPIC96, PSAVERT, TDSP, REVOLSL, DSPI, GASREGW, and USREC. Monthly values are within-month averages when higher-frequency data are pulled, and annual

values are within-year averages. CPI inflation is year-over-year monthly CPI inflation. Rolling inflation volatility is the rolling 12-month standard deviation of monthly CPI inflation. Real disposable-income growth is year-over-year real disposable personal income growth. Revolving credit is divided by 1,000 before normalization by disposable personal income because REVOLSL is reported in millions while DSPI is reported in billions. The replication workflow rebuilds tables, figures, and processed data from public source series.